

SAEED AHMED KHAN

PH.D. • SENIOR MEMBER IEEE

Dean of Engineering & Technology | Researcher in Flexible Electronics,
Sensors and Energy Harvesting

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◆ [Google Scholar](#) | [Scopus](#)

PROFESSIONAL PROFILE

A senior academic and researcher with **18+ years** of combined experience across higher education, applied research, and engineering practice in **five countries** (Pakistan, China, South Korea, Croatia/EU, and trans-regional collaborations). Currently serving as **Dean of the Faculty of Engineering and Technology** at Sukkur IBA University, with prior leadership as Director of the Quality Enhancement Cell. My research bridges nanomaterials, flexible electronics, and self-powered sensing — with **~1,000 citations** across publications in *ACS Sensors*, *Nano Energy*, *Chemical Engineering Journal*, *Journal of Materials Chemistry C*, and Wiley/Elsevier flagship venues. I bring an Erasmus Mundus background, an HEC-funded research portfolio, and a track record of building accreditation, quality-assurance, and graduate-supervision systems aligned with European higher-education standards.

AT A GLANCE

~1,000 GOOGLE SCHOLAR CITATIONS	60+ PEER-REVIEWED PUBLICATIONS	18+ YEARS IN ACADEMIA & INDUSTRY	5 COUNTRIES OF RESEARCH PRACTICE	9 THESES SUPERVISED (UG/MS/PHD)	30+ JOURNALS & CONFERENCES REVIEWED
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RESEARCH FOCUS

Flexible & Wearable Electronics	Self-Powered Sensors	Triboelectric Nanogenerators	Thermogalvanic Hydrogels
Energy Harvesting & Conversion	Carbon Nanomaterials & Thin Films	Bio-signal & Health Monitoring	Printed & Soft Electronics

ACADEMIC & PROFESSIONAL EXPERIENCE

Dean, Faculty of Engineering and Technology Sukkur IBA University • Pakistan

JANUARY 2026 – PRESENT

- Provide academic and strategic leadership for the Faculty, overseeing curriculum, research, and faculty development across electrical, computer, and allied engineering programs.
- Drive internationalization, accreditation, and research-grant strategy in alignment with HEC and global engineering-education standards.

Director, Quality Enhancement Cell (QEC) Sukkur IBA University • Pakistan

JUNE 2023 – JANUARY 2026

- National Certified Reviewer of Higher Education Institutions (Sindh HEC) — credentialed via 100-hour formal review training.
- Designed and implemented institutional quality-assurance frameworks: program self-assessment, accreditation audits, and continuous-improvement cycles.
- Led the institution's submissions for national HEC rankings and international rankings including THE Impact Rankings, THE World Ranking, and UI GreenMetric — building data pipelines and evidence portfolios.
- Established faculty/staff training programs on Outcome-Based Education (OBE), academic integrity, and quality benchmarking.

Research Professor / Postdoctoral Fellow *Sungkyunkwan University (SKKU) · South Korea*

MAY 2022 – APRIL 2023

- ▶ Conducted advanced research on electrohydrodynamic (EHD) jet printing, flexible PDMS-based devices, and triboelectric nanogenerators with the School of Mechanical Engineering.
- ▶ Co-authored multiple high-impact publications in Polymers, Energy Reports, Heliyon, and Sensors and Actuators A — strengthening cross-border research linkages with East Asian and European groups.

Associate Professor *Sukkur IBA University · Pakistan*

AUGUST 2019 – JANUARY 2026

- ▶ Member of the University Syndicate (statutory body) for three years from October 2021.
- ▶ Supervised undergraduate, M.Eng., and Ph.D. research; built laboratory capability in flexible-sensor fabrication and characterization.
- ▶ Served as Chief Editor, Sukkur IBA Journal of Emerging Technologies (HEC Y-category, 2018–2022).

Assistant Professor *Sukkur IBA University · Pakistan*

JULY 2012 – AUGUST 2019

- ▶ Designed and delivered core electrical / electronics engineering courses; established research direction in flexible sensors and nanomaterials.
- ▶ Co-organized and chaired iCOMET — the International Conference on Computing, Mathematics and Engineering Technologies (1st, 2nd, 3rd editions).

Lecturer *ISRA University · Hyderabad, Pakistan*

JANUARY 2012 – JULY 2012

Research Assistant *University of Zagreb · Croatia (EU)*

SEPTEMBER 2010 – JULY 2011

- ▶ Erasmus Mundus exchange — research on advanced silicon bipolar transistor characterization.

Industrial Roles — Instrumentation & Electrical Engineering *Habib Sugar Mills · Government of Pakistan · Khairpur Sugar Mills · Pakistan*

JANUARY 2006 – MAY 2009

- ▶ Hands-on industrial engineering — instrumentation, electrical maintenance, and works management — providing applied grounding for later academic research on sensors and energy systems.

EDUCATION

Doctor of Philosophy (Ph.D.) **2017**

University of Electronic Science and Technology of China (UESTC) · Chengdu, P.R. China

Thesis: *The Design and Construction of Flexible Sensors Based on Carbon Nanomaterials*

✚ Awarded under the Chinese Government Scholarship.

Master of Engineering — Electronic Systems Engineering **2011**

Mehran University of Engineering & Technology, Jamshoro · Pakistan · Research stay: University of Zagreb, Croatia (EU)

Thesis: *Electrical Properties of Fabricated Transistor Regions in Advanced Silicon Bipolar Technologies*

✚ Erasmus Mundus Scholarship recipient — University of Zagreb, Croatia.

Bachelor of Engineering — Electronics Engineering **2005**

Mehran University of Engineering & Technology, Jamshoro · Pakistan

Thesis: *Image Edge Detection and Object Recognition Using Wavelet Transform*

RESEARCH GRANTS & FUNDED PROJECTS

2024

PI

Sindh Research Support Programme (SRSP) · *Principal Investigator*

Awarded PKR 2.4 million by the Sindh Higher Education Commission for advanced sensor and materials research.

2019

CO-PI

Start-up Research Grant (SRGP), HEC Pakistan · *Co-Principal Investigator*

Brain–computer interface project on EEG signal processing to support paralysis patients.

2018

PI

Start-up Research Grant (SRGP), HEC Pakistan · *Principal Investigator*

Flexible sensors based on nanomaterials — establishing the institutional research line.

2013

INV

ICT R&D Fund — Sustainable Energy Project · *Investigator*

Designed an alternative-energy system for street-lighting and residential power, reducing dependence on thermal generation.

SELECTED PUBLICATIONS

Curated selection from 60+ peer-reviewed publications. Full list on [Google Scholar](#) and [Scopus](#).

RECENT HIGH-IMPACT (2024 – 2025)

1. Y. Li, Y. Zhang, B. Chang, S.A. Khan, X. Cui, J. Zhang, H. Zhang. "Piezoionic Effect in Gels: Multifunctional Applications from Energy Harvesting to Intelligent Sensing." *Nanoscale*, 2025. <https://doi.org/10.1039/D5NR04389A>
2. C. Bai, Y. Yao, S.A. Khan, J. Li, F. Yi, J. Zhang, H. Shi, H. Zhang. "Configurational Entropy-Change-Driven Enhancement of Thermoelectric Performance in PVA-Based Hydrogel Thermogalvanic Cells." *Journal of Power Sources*, Vol. 665, 239060, 2025. <https://doi.org/10.1016/j.jpowsour.2025.239060>
3. R. Wei, X. Cui, S.A. Khan, C. Bai, et al.. "Self-Powered Versatile On-Hand Virtual Interface Enabled by Thermogalvanic Hydrogel." *Chemical Engineering Journal*, 167739, 2025. <https://doi.org/10.1016/j.cej.2025.167739>
4. Y. Yao, S.A. Khan, X. Cui, K. Liu, S. Wen, H. Zhang. "Hydrogel-Based Smart Contact Lens for Enhanced Functional Applications." *Biomedical Instrumentation*, 100003, 2025. <https://doi.org/10.1016/j.bmi.2025.100003>
5. X. Cui, Y. Nie, S.A. Khan, et al.. "Thermoelectric Gel Enabling Self-Powered Facial Perception for Expression Recognition and Health Monitoring." *ACS Sensors*, 2024. <https://doi.org/10.1021/acssensors.4c03099>
6. Y. Zhang, H. Wang, S.A. Khan, J. Li, C. Bai, H. Zhang, R. Guo. "Deep-Learning-Assisted Thermogalvanic Hydrogel Fiber Sensor for Self-Powered In-Nostril Respiratory Monitoring." *Journal of Colloid and Interface Science*, 2024. <https://doi.org/10.1016/j.jcis.2024.09.132>
7. S.A. Khan, S. Ali, J. Moon, A. Ali, R. ul Hassan, D.H. Cho, D. Byun. "Multi-Energy Harvesting: Integrating Contact-Mode and Slide-Mode Triboelectric Nanogenerators, and Solar Technologies for Efficient Power Generation in Small Electronics." *Energy Reports*, Vol. 12, 4232–4240, 2024. <https://doi.org/10.1016/j.egyr.2024.09.034>
8. S. Sang, C. Bai, W. Wang, S.A. Khan, et al.. "Finger-Temperature-Driven Thermogalvanic Gel-Based Smart Pen for Identity Recognition, Stroke Analysis, and Grip-Posture Assessment." *Nano Energy*, 2024. <https://doi.org/10.1016/j.nanoen.2024.109366>
9. C. Tian, S.A. Khan, Z. Zhang, X. Cui, H. Zhang. "Thermoelectric Hydrogel Electronic Skin for Passive Multimodal Physiological Perception." *ACS Sensors*, 2024. <https://doi.org/10.1021/acssensors.3c02172>
10. X. Yang, Z. Zhang, S.A. Khan, et al.. "Thermogalvanic Organohydrogel-Based Non-Contact Self-Powered Electronics for Advancing Smart Agriculture." *Journal of Materials Chemistry C*, 2024. <https://doi.org/10.1039/D3TC04133F>

FOUNDATIONAL & HIGH-CITED WORK (2017 – 2023)

11. J. Li, Z. Wang, S.A. Khan, N. Li, Z. Huang, H. Zhang. "Self-Powered Information Conversion Based on Thermogalvanic Hydrogel with Interpenetrating Networks for Nursing Aphasic Patients." *Nano Energy*, Vol. 113, 108612, 2023. <https://doi.org/10.1016/j.nanoen.2023.108612>
12. H. Yang, S.A. Khan, N. Li, R. Fang, Z. Huang, H. Zhang. "Thermogalvanic Gel Patch for Self-Powered Human-Motion Recognition Enabled by Photothermal–Electric Conversion." *Chemical Engineering Journal*, 145247, 2023. <https://doi.org/10.1016/j.cej.2023.145247>

13. X. Li, J. Li, T. Wang, S.A. Khan, Z. Yuan, Y. Yin, H. Zhang. "Self-Powered Respiratory Monitoring Strategy Based on Adaptive Dual-Network Thermogalvanic Hydrogels." *ACS Applied Materials & Interfaces*, Vol. 14, 43, 48743–48751, 2022. <https://pubs.acs.org/doi/full/10.1021/acsami.2c14239>
14. R.U. Hassan, S.M. Khalil, S.A. Khan, S. Ali, J. Moon, D.H. Cho, D. Byun. "High-Resolution, Transparent, and Flexible Printing of Polydimethylsiloxane via Electrohydrodynamic Jet Printing for Conductive Electronic Devices." *Polymers*, Vol. 14(20), 4373, 2022. <https://doi.org/10.3390/polym14204373>
15. H. Khan, A.M. Soomro, S.A. Khan, et al.. "Highly Sensitive Mechano-Optical Strain Sensor Based on 2D Material for Human Quality-Gait Monitoring and High-End Robotic Applications." *Journal of Materials Chemistry C*, 2021. <https://doi.org/10.1039/D1TC03519C>
16. X. Xie, H. Zhang, S.A. Khan, et al.. "A Movable-Electrode Triboelectric Nanogenerator Fabricated Using a Pencil Lead for Self-Powered Locating-Collision." *Advanced Engineering Materials*, Vol. 22(6), 2000109, 2020. <https://onlinelibrary.wiley.com/doi/full/10.1002/adem.202000109>
17. S. Cao, R. Guo, S.A. Khan, et al.. "Building Self-Powered Emergency Electronics Based on Hybrid Nanogenerators for Field Survival/Rescue." *Energy Science & Engineering*, Vol. 7(3), 574–581, 2019. <https://onlinelibrary.wiley.com/doi/full/10.1002/ese3.497>
18. Y. Xie, H. Zhang, G. Yao, S.A. Khan, M. Gao, Y. Su, W. Yang, Y. Lin. "Intelligent Sensing System Based on Hybrid Nanogenerator by Harvesting Multiple Clean Energy." *Advanced Engineering Materials*, Vol. 20(1), 1700886, 2018. <https://onlinelibrary.wiley.com/doi/full/10.1002/adem.201700886>
19. S.A. Khan, H.L. Zhang, Y. Xie, M. Gao, M.A. Shah, A. Qadir, Y. Lin. "Flexible Triboelectric Nanogenerator Based on Carbon Nanotubes for Self-Powered Weighing." *Advanced Engineering Materials*, Vol. 19(3), 1600710, 2017. <https://onlinelibrary.wiley.com/doi/full/10.1002/adem.201600710>
20. Z. Huang, M. Gao, Z. Yan, T. Pan, S.A. Khan, Y. Zhang, H. Zhang, Y. Lin. "Pyramid Microstructure with Single-Walled Carbon Nanotubes for Flexible and Transparent Micro-Pressure Sensor with Ultra-High Sensitivity." *Sensors and Actuators A: Physical*, Vol. 266, 345–351, 2017. <https://www.sciencedirect.com/science/article/abs/pii/S0924424717308543>
21. S.A. Khan, M. Gao, Y. Zhu, Z. Yan, Y. Lin. "MWCNTs-Based Flexible and Stretchable Strain Sensors." *Journal of Semiconductors*, Vol. 38(5), 053003, 2017. <https://iopscience.iop.org/article/10.1088/1674-4926/38/5/053003/meta>

BOOK CHAPTER

22. S.A. Khan, Shamsuddin, A.Q. Rahimoon, A. Abro. "Powering Healthcare IoT Sensors Based on Triboelectric Nanogenerator and Its Applications." *IGI Global*, 2019. <https://www.igi-global.com/chapter/powering-healthcare-iot-sensors-based-triboelectric-nanogenerator/243908>

EDITORIAL SERVICE & PEER REVIEW

Chief Editor *Sukkur IBA Journal of Emerging Technologies (SJET)* — HEC-recognized (Y category), 2018–2022.

General Chair / Co-General Chair *International Conference on Computing, Mathematics and Engineering Technologies (iCoMET)* — 1st, 2nd, and 3rd editions.

Fellowship Reviewer *Paris Region Fellowship Programme (ParisRegionFP)* — international postdoctoral fellowship evaluation, France.

Reviewer for leading journals (selected):

<i>Advanced Materials</i>	<i>Advanced Functional Materials</i>	<i>Advanced Materials Interfaces</i>
<i>Nano-Micro Letters</i>	<i>Small</i>	<i>NANO</i>
<i>ACS Sensors</i>	<i>Sensors</i>	<i>Journal of Materials Chemistry A</i>
<i>Journal of Materials Chemistry C</i>	<i>Journal of Applied Energy</i>	<i>Energy & Environmental Materials</i>
<i>International Journal of Energy Research</i>	<i>IEEE Access</i>	<i>IEEE Trans. Intelligent Transportation Systems</i>
<i>IEEE/ASME Trans. Mechatronics</i>	<i>ECS Journal of Solid State Science and Technology</i>	<i>Materials</i>
<i>Applied System Innovation</i>	<i>Electronics</i>	<i>Materials Performance and Characterization</i>
<i>International Journal of Imaging Systems and Technology</i>		

TEACHING & RESEARCH SUPERVISION

SUPERVISION

- ▶ Supervised 6 undergraduate research theses (B.E. Electronic / Electrical Engineering).
- ▶ Supervised 3 Master of Engineering theses, with publications in indexed venues.

COURSES DESIGNED AND DELIVERED (SELECTED)

▶ VLSI Technology	▶ RF MEMS	▶ Sensors Technology
▶ Instrumentation & Measurement	▶ Linear Control Systems	▶ Advanced Control Engineering
▶ Electronic Devices and Circuits	▶ Linear / Network Circuit Analysis	▶ Electronic Circuit Design
▶ Optoelectronics	▶ Introduction to Nanotechnology	▶ New Energy Materials & Devices
▶ Electronic Packaging	▶ Analog & Digital Electronics	▶ Analog & Digital Communication
▶ Signals & Systems	▶ Programming (MATLAB · Python · C++)	▶ Human-Computer Interaction

ACADEMIC GOVERNANCE & COMMITTEES (SELECTED)

- ▶ Member, University Syndicate (statutory body), Sukkur IBA University — three-year term from October 2021.
- ▶ Member, Board of Studies, Department of Electronics, University of Sindh, Jamshoro.
- ▶ Member, Board of Faculty Council, Benazir Bhutto University of Technology & Skill Development.
- ▶ Member, OBE Accreditation Committee — curriculum review for undergraduate and postgraduate programs.
- ▶ Member, Plagiarism Committee, Board of Faculty, Academic Council, BASR, Syndicate, and Senate of Sukkur IBA University.
- ▶ Member, Program Review Committee (QEC) — assessment of M.E./M.S./Ph.D. self-assessment reports.
- ▶ Member, Departmental Strategic Committee — review of academic and co-curricular activities.
- ▶ Subject Expert (Electronics) — selection committees for academic and research positions.

HONORS & AWARDS

2024 Sindh HEC Sindh Research Support Programme Grant

PKR 2.4 million as Principal Investigator.

2019 HEC Pakistan Start-up Research Grant

Co-Principal Investigator.

2018 HEC Pakistan Start-up Research Grant

Principal Investigator.

2014 Chinese Government Scholarship

Full Ph.D. fellowship at UESTC, P.R. China.

2010 Erasmus Mundus Scholarship (EU)

Master of Engineering programme at the University of Zagreb, Croatia.

TECHNICAL SKILLS, MEMBERSHIPS & LANGUAGES

TECHNICAL & SOFTWARE	MEMBERSHIPS	LANGUAGES
PROGRAMMING MATLAB · LabVIEW · Python · C++ SIMULATION & FEA ANSYS · COMSOL CAD SolidWorks · AutoCAD · Origin · EHD CHARACTERIZATION & SYNTHESIS CVD · EB-PVD · SEM · XRD · Semiconductor Parameter Analyzer · 3D Printing (inkjet EHD)	IEEE Senior Member (since 2012) PAKISTAN ENGINEERING COUNCIL Member (since 2006) HEC PAKISTAN National Certified Reviewer of Higher Education Institutions (Sindh HEC, 100- hour training).	ENGLISH Fluent (academic and professional) URDU / SINDHI Native CROATIAN / MANDARIN CHINESE Beginner — gained through long-term residence and study in Croatia and China.

REFERENCES

Prof. Lin Yuan

Professor, School of Microelectronics & Solid-State Electronics

University of Electronic Science and Technology of China (UESTC)

P.R. China

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Prof. Doyoung Byun

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Sungkyunkwan University (SKKU)

South Korea

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Prof. Hulin Zhang

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